# SHTC equations with Material Metric

## Workshop on Discrete Exterior Calculus 2026

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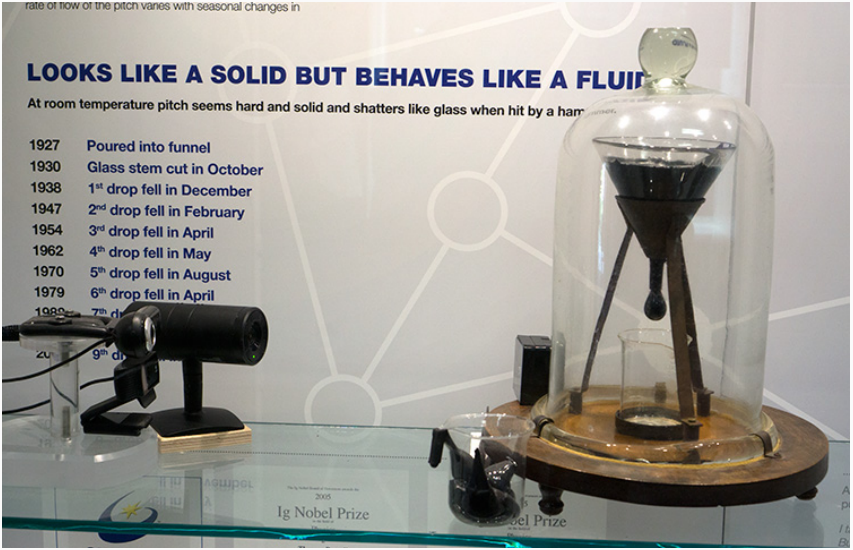
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**Question**: Can we construct a continuum theory that can handle the fluid-solid spectrum continuously — without switching frameworks at an arbitrary threshold?




Source: https://www.theandytchannel.com/wp-content/uploads/2019/08/pitch7-1.jpg

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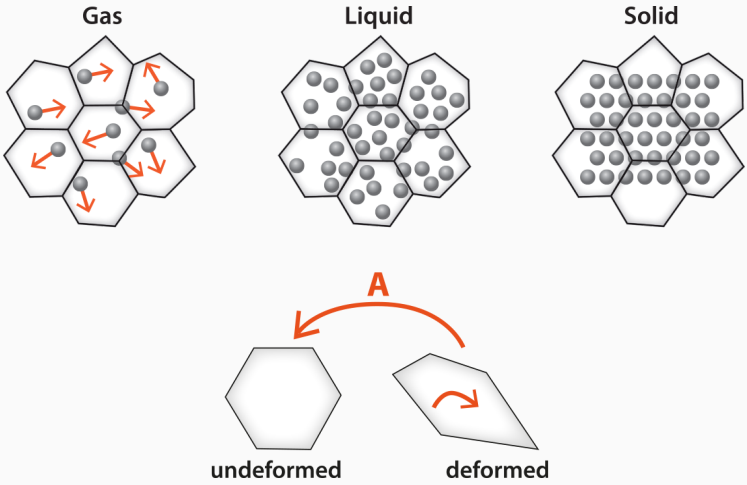
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The fluidity is introduced via relaxation of distortion field $\mathbf{A}$.

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Continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \tag{1}$$

Momentum equation:

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\mathbf{v} \otimes \rho \mathbf{v} - \mathbf{S}) = 0 \tag{2}$$

Equation for distortion:

$$\frac{\partial \mathbf{A}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{A} + \mathbf{A} \nabla \mathbf{v} = -\frac{1}{\theta} E_{\mathbf{A}} \tag{3}$$

Equation for energy:

$$\frac{\partial \rho E}{\partial t} + \nabla \cdot (\mathbf{v} \rho E - \mathbf{S} \mathbf{v}) = 0 \tag{4}$$

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$$\frac{dA}{dt} = \{A,\, H\} + \left\langle A_{\mathsf{x}},\, \frac{\delta \Xi}{\delta S_{\mathsf{x}}} \right\rangle$$

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Plasticity traditionally:

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**Question**: What if plasticity is not a decomposition of deformation, but a change in the material itself?

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**Proposition**: Treat the material body as a Riemannian manifold $(\mathcal{M}, \mathbf{g})$.

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This yields a clean conceptual separation from classical SHTC model.
In **Metric SHTC** we have:

| Variable | Meaning |
|---|---|
| $\mathbf{A}$ | Kinematic mapping |
| $\mathbf{g}$ | Material internal state |

# Two channels of dissipation

**Dissipation in distortion**

Classical SHTC:

$$\partial_t \mathbf{A} = -\mathbf{A}\mathbf{L} - \frac{1}{\theta} E_{\mathbf{A}}$$

Distortion loses memory of the reference mapping. Describes viscous or plastic flow.

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**Dissipation in material metric**

Metric SHTC:

$$\partial_t \mathbf{g} = -v \cdot \nabla \mathbf{g} - \frac{1}{\theta} E_{\mathbf{g}}$$

The natural microstructure evolves irreversibly. Describes internal state changes.

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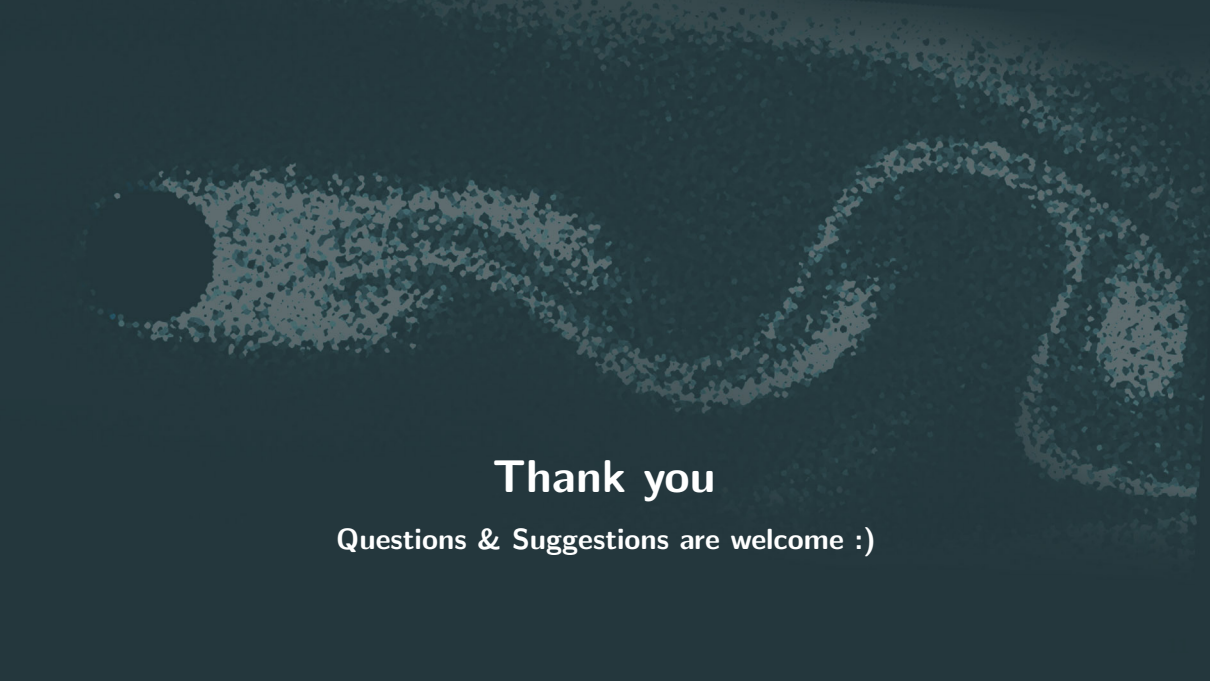
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GENERIC splitting is preserved by design


Thank you
Questions & Suggestions are welcome :)